

Dexamethasone facilitates lipid accumulation in chicken skeletal muscle.

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The effects of glucocorticoid on lipid metabolism of broiler chicken (*Gallus gallus domesticus*) skeletal muscle were investigated. Male Arbor Acres chickens (35 days old) were subjected to dexamethasone treatment for 3 days. We found that dexamethasone retards body growth while facilitating lipid accumulation. In *M. pectoralis major* (PM), dexamethasone increased the expression of glucocorticoid receptor (GR), fatty acid transport protein 1 (FATP1), heart fatty acid-binding protein (H-FABP) and long-chain acyl-CoA dehydrogenase (LCAD) mRNA and decreased the expression of liver carnitine palmitoyltransferase 1 (L-CPT1), adenosine-monophosphate-activated protein kinase (AMPK) $\alpha 2$ and lipoprotein lipase (LPL) mRNA. LPL activity was also decreased. In *M. biceps femoris* (BF), the levels of GR, FATP1 and L-CPT1 mRNA were increased. AMPK α (Thr172) phosphorylation and CTP1 activity of skeletal muscle were decreased by dexamethasone. In fed chickens, dexamethasone enhanced very low-density lipoprotein receptor (VLDLR) expression and AMPK activity in muscle, but it impaired the expression of LPL and L-CPT1 mRNA and LPL activity in PM and augmented the expression of GR, LPL, H-FABP, L-CPT1, LCAD and AMPK $\alpha 2$ mRNA in BF. Adipose triglyceride lipase (ATGL) protein expression was not affected by dexamethasone. In conclusion, in the fasting state, dexamethasone-induced-retarded fatty acid utilisation may be involved in the augmented intramyocellular lipid accumulation in both glycolytic (PM) and oxidative (BF) muscle tissues. In the fed state, dexamethasone promoted the transcriptional activity of genes related to lipid uptake and oxidation in muscles. Unmatched lipid uptake and utilisation are suggested to be involved in the augmented intramyocellular lipid accumulation.